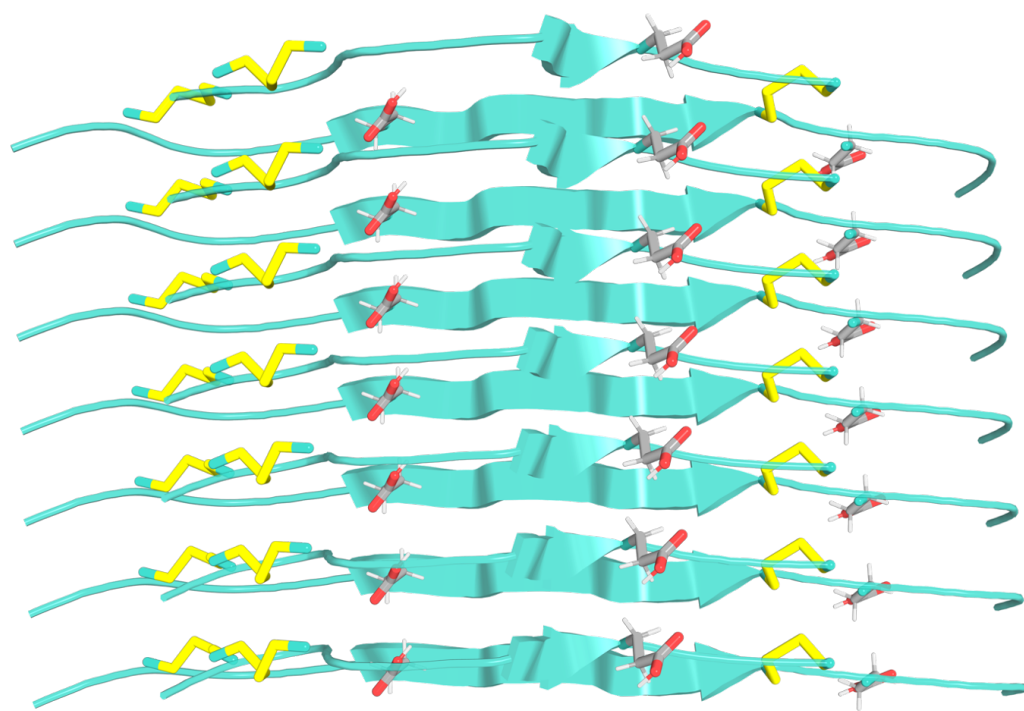


pH-Dependent Free Energies of Insulin Fibrillation from Protonation Ensemble Statistics

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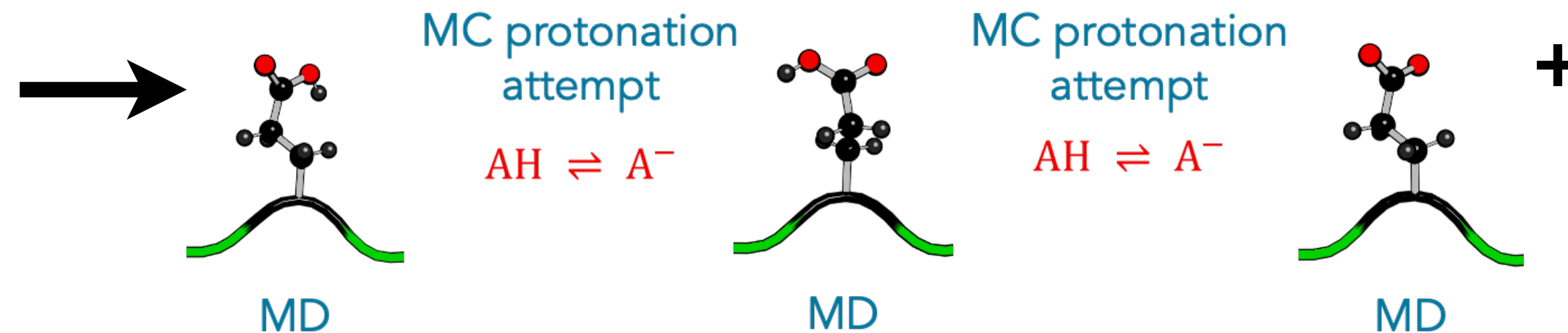
Motivation

What is the origin of pH-induced fibrillation in insulin?



Simulation Method

We treat the protonation states of amino acids as a fluctuating degree of freedom



Free Energy Calculation Method

Rather than using TI, we construct a statistical optimal method for pH-dependent free energy calculations

$$\sum_{N=0}^{N_{\max}} \frac{H_{\text{tot}}(N) \exp[-\beta \Upsilon_m + \beta \mu_{\text{H}^+}^m N]}{\sum_{j=1}^K M_j \exp[-\beta \Upsilon_j + \beta \mu_{\text{H}^+}^j N]} = 1$$
$$\forall m = 1, \dots, K$$

Results & Discussion

$$\underbrace{(\text{p}\bar{K}_{a,1}, \text{p}\bar{K}_{a,2})}_{\text{macroscopic}} \rightarrow \underbrace{(\text{p}K_{a,1}, \text{p}K_{a,2}, W)}_{\text{microscopic}}$$

$$0 \leq W \leq W_{\max}$$

